

EXPERIMENT 25

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Aim

To write a menu-driven C program using a **switch** statement and nested loops to display four decreasing patterns. Each pattern is implemented in a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display a menu containing the four decreasing-pattern choices.
3. Read the user's menu choice.
4. Use a switch statement to call the corresponding parameterless void function.
5. Inside the selected function, read the number of rows n.
6. Use nested loops to generate the selected decreasing pattern.
7. If the choice is outside 1-4, print an invalid-choice message.
8. Stop.

C Program

```
#include <stdio.h>

void decreasingStar(void);
void decreasingNatural(void);
void decreasingRepeated(void);
void reverseNumber(void);

int main(void) {
    int choice;

    printf("DECREASING PATTERNS MENU\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");
    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: decreasingStar(); break;
        case 2: decreasingNatural(); break;
        case 3: decreasingRepeated(); break;
        case 4: reverseNumber(); break;
        default: printf("Invalid choice!\n");
    }
    return 0;
}

void decreasingStar(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) printf("* ");
        printf("\n");
    }
}

void decreasingNatural(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}
```

```

void decreasingRepeated(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}

void reverseNumber(void) {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = i; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

```

Test-Case Table

Test	Choice	n	Expected output
1	1	4	**** / *** / ** / *
2	2	4	1 2 3 4 / 1 2 3 / 1 2 / 1
3	3	4	4 4 4 4 / 3 3 3 / 2 2 / 1
4	4	4	4 3 2 1 / 3 2 1 / 2 1 / 1
5	7	-	Invalid choice!

Sample Execution (n = 4)

Choice 1 - Decreasing right-angled triangle

```

* * * *
* * *
* *
*

```

Choice 2 - Decreasing natural-number pattern

```

1 2 3 4
1 2 3
1 2
1

```

Choice 3 - Decreasing repeated-number pattern

```

4 4 4 4
3 3 3
2 2
1

```

Choice 4 - Reverse-number triangle

```

4 3 2 1
3 2 1
2 1
1

```

Result

The program successfully uses a switch statement to select one of four decreasing patterns. Each pattern is produced by a separate parameterless void function using nested loops, and the prepared test cases verify all menu

options and an invalid choice.